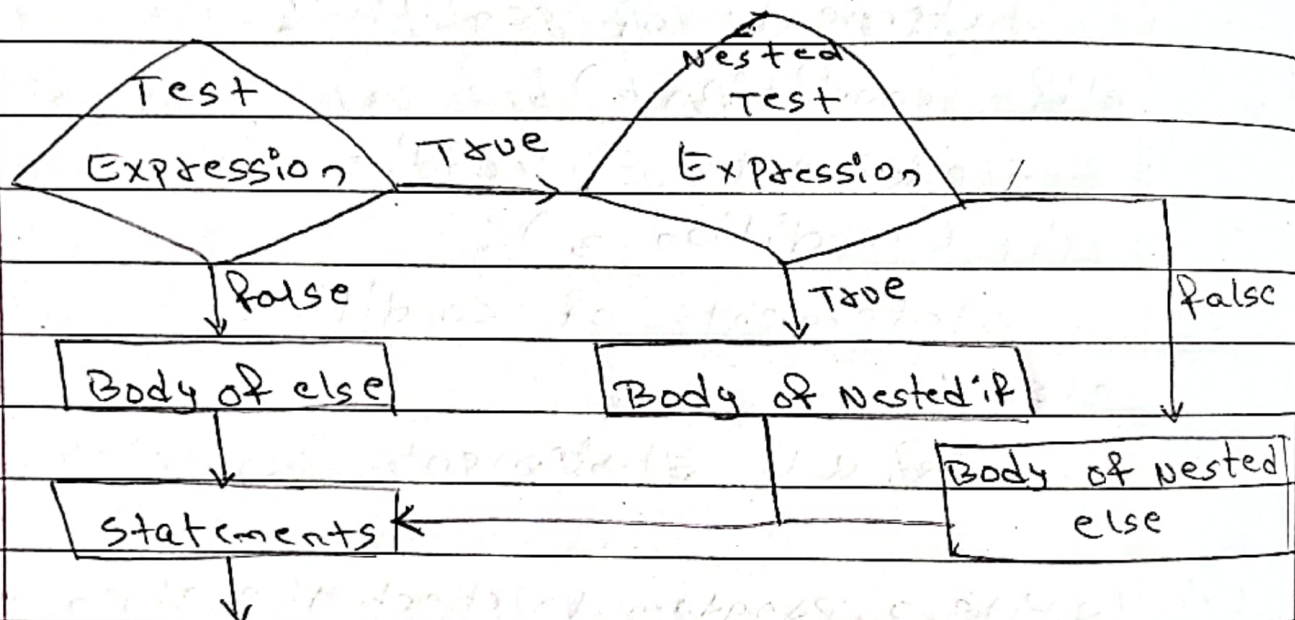


④ Nested if

⇒ It is used to place one if statement inside another if statement. This helps when we need to check multiple conditions in a structured way.

• Flowchart



• Syntax

```

if (Condition 1) : # Outer if
    if (Condition 2) : # Inner if
        statements of inner if
    else : # Inner else
        statements of inner else
else :
    statements of outer else
  
```

Ex: age = 14
country = "india"
if (country == "india" or age >= 18):
 if (country == "india"):
 if (age >= 18):
 print("You are eligible to vote")
 else:
 print("You are not eligible because your
 age is not more than 18")
 else:
 print("You are not eligible to
 vote bcz invalid country")
else:
 print("You entered both values invalid")

O/P You are not eligible because your age
is not more than 18

Ex: num = int(input())
if (num >= 0):
 if (num == 0):
 print("zero")
 else:
 print("positive")
else:

print("Negative")

O/P

-89

76

-89 is Negative

76 is Positive

Ex 3: Triangle type checker take three sides of a triangle as input and check it is equilateral (a, b, c should equal), isosceles (two sides should equal), scalene (All three sides are different) or not triangle (sum of any two sides > other side).

Program

a = 15

b = 15

c = 20

if (a+b > c and a+c > b and b+c > a):

if (a == b == c):

print("Equilateral triangle")

elif (a == b or b == c or c == a):

print("isosceles triangle")

else:

print("scalene triangle")

else:

print("Its not triangle")

O/P

isosceles triangle